

The Frontier Occurrence-Lift Contract: Field Descent, Lineage Completeness, and Schema Nonidentifiability

Liang Wang
School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

The complete TPC cut archive reaches every eligible-tail and frontier path but does not yet reach the canonical determinant, ordered zero-mode, or physical occurrence archives. We identify the common missing object: a conservative row-separated occurrence lift

$$L_X : \mathbb{C}^{\mathcal{C}_X^{\text{ns}}} \longrightarrow \mathbb{C}^{\mathcal{O}_X}.$$

One cut path may have several downstream occurrences, so this object is a sparse matrix rather than a single-valued label map. We prove a field-descent criterion: a downstream label is determined by the current cut schema precisely when it is constant on every fiber of the forgetful completion map. The inherited cut-stage selector is therefore exactly $P_{h_0}^{\text{cut}} = I$, but determinant bins, zero-mode order, physical groups, and the downstream shift selector do not descend from the current fields. This is a scoped nonidentifiability theorem for the present archive, not an impossibility result for an augmented construction. A deterministic audit binds every nonsoft cut path to an explicit lift obligation. The committed finite sample has 2988 frontier paths and no eligible-tail path; that finite emptiness is not promoted to asymptotic totality. The new result is structural L0/L1 progress and supplies no positive fixed-shift arithmetic saving.

1 The exact point reached by the cut archive

Fix the literal hard-packet scope

$$\mathfrak{s} = (X, h_0, Q, U, V, W_{\text{id}}, \nu_{\text{phys}})$$

and a native tuple (ℓ, k, d) . TPC-133 enumerates every tuple in its certified support envelope; TPC-134 expands it into exact dyadic paths; and TPC-135–136 assign one of the terminal types

$$\text{EPS}, \quad \text{ETO}, \quad \text{FUM}$$

without deleting a path [3–6]. They stand for eligible-prefix-soft, eligible-tail-open and frontier-unmapped. Put

$$\mathcal{C}_X^{\text{ns}} = \mathcal{C}_X^{\text{ETO}} \sqcup \mathcal{C}_X^{\text{FUM}}.$$

This is the required domain for every downstream map in TPC-136. Calling only the FUM rows “the domain” would silently delete the eligible tail.

The cut record stores its exact path identifiers, packet scope, native tuple, dyadic block, cutoff D_0 , terminal reason, multiplier expression, h_0 , and physical normalization. Joining on the exact scope and tuple recovers the TPC-133 coefficient expression. A SHA-256 value checks file integrity; it is not the mathematical identity.

Definition 1.1 (Three support roles). The *formal support envelope* is the enumerated domain on which the symbolic packet is defined. A *canonical parent carrier* is obtained only after a theorem-backed inverse aggregation. The *actual active support* consists of the resulting records whose literal coefficients are certified nonzero. These roles are different typed sets.

The current cut archive is an envelope archive. A symbolic factor such as $r_Q(\ell k + h_0)$ may vanish. Conversely, an unevaluated coefficient is not an exact-zero certificate. We therefore retain every formal nonsoft path when testing map totality.

2 The missing occurrence lift

Definition 2.1 (Fully typed downstream occurrence). A downstream occurrence $o \in \mathcal{O}_X$ contains:

- (i) an occurrence identifier, source and target record identifiers, stage identifier, and exact edge multiplier;
- (ii) the exact cut-path, native-tuple and packet-scope lineage;
- (iii) source and target shift tags and one inherited physical normalization;
- (iv) all content, mask, polarization, boundary and collision tags;
- (v) canonical-parent, determinant-fiber and zero-mode fields when the corresponding maps are invoked; and
- (vi) physical group, cover, reconstruction and occurrence-registry fields when physical reassembly is invoked.

Definition 2.2 (Occurrence lift). An occurrence lift is an exact sparse matrix

$$L_X : \mathbb{C}^{\mathcal{C}_X^{\text{ns}}} \longrightarrow \mathbb{C}^{\mathcal{O}_X}.$$

It is *row separated*: separate transformation paths remain separate rows even if they end at the same mathematical occurrence. It is *conservative* when

$$\mathbf{1}_{\mathcal{O}_X}^T L_X = \mathbf{1}_{\mathcal{C}_X^{\text{ns}}}^T. \quad (1)$$

It is *metadata-faithful* when it intertwines the native tuple, h_0 , and ν_{phys} on every nonzero edge.

One cut path can split under polarization, content or physical reconstruction. Hence L_X is not assumed to be a function. Its column sum can be one even when its individual entries are signed or complex. Equation (1) is a coefficientwise identity, not a bound on the associated arithmetic scalar.

Proposition 2.3 (Exact extension of cut reconnection). Let $\mathcal{N}_X$ be the native-record set and let M_X^{cut} be the conservative TPC-136 cut matrix. After ordering its rows by terminal type, write

$$M_X^{\text{cut}} = \begin{pmatrix} M_X^{\text{EPS}} \\ M_X^{\text{ns}} \end{pmatrix}, \quad M_X^{\text{ns}} : \mathbb{C}^{\mathcal{N}_X} \longrightarrow \mathbb{C}^{\mathcal{C}_X^{\text{ns}}}.$$

If (1) holds, then

$$\mathbf{1}_{\text{EPS}}^T M_X^{\text{EPS}} + \mathbf{1}_{\mathcal{O}_X}^T L_X M_X^{\text{ns}} = \mathbf{1}_{\mathcal{N}_X}^T. \quad (2)$$

Equivalently, adjoining the identity on the already soft rows gives $\hat{L}_X = I_{\text{EPS}} \oplus L_X$ and $\mathbf{1}^T \hat{L}_X M_X^{\text{cut}} = \mathbf{1}^T$. Thus the extended archive reconnects every native coefficient exactly. This conclusion neither identifies a physical occurrence nor estimates any nonsoft sum.

Proof. Multiply $\mathbf{1}_{\mathcal{O}_X}^T L_X = \mathbf{1}_{\mathcal{C}_X^{\text{ns}}}^T$ by M_X^{ns} , add the unchanged EPS contribution, and use the full TPC-136 column identity $\mathbf{1}_{\text{EPS}}^T M_X^{\text{EPS}} + \mathbf{1}_{\mathcal{C}_X^{\text{ns}}}^T M_X^{\text{ns}} = \mathbf{1}_{\mathcal{N}_X}^T$. $\square$

3 Field descent and nonidentifiability

Let $\widetilde{\mathcal{O}}_{X\text{schema}}$ be the maximal formal completion class constrained only by the fields and algebraic identities explicitly archived in the present contracts. In particular, it does not silently impose an unarchived actual-support condition. We include at least one formal completion above every cut row, and let

$$\pi : \widetilde{\mathcal{O}}_{X\text{schema}} \longrightarrow \mathcal{C}_X^{\text{ns}}$$

be the resulting surjective map that forgets every field absent from the current schema.

Theorem 3.1 (Field-descent criterion). *For a set Y and a downstream label $\lambda : \widetilde{\mathcal{O}}_{X\text{schema}} \rightarrow Y$, the following are equivalent:*

- (a) *there exists a label $\bar{\lambda} : \mathcal{C}_X^{\text{ns}} \rightarrow Y$ satisfying $\lambda = \bar{\lambda} \circ \pi$;*
- (b) *λ is constant on every fiber $\pi^{-1}(c)$.*

The descended label $\bar{\lambda}$ is unique.

Proof. If $\lambda = \bar{\lambda} \circ \pi$, equal images under π have equal λ -values. Conversely, define $\bar{\lambda}(c) = \lambda(o)$ for any $o \in \pi^{-1}(c)$. Fiber constancy makes the definition independent of the choice. Surjectivity gives both existence on all cut records and uniqueness. $\square$

Proposition 3.2 (Schema-completion labels do not descend). *On $\widetilde{\mathcal{O}}_{X\text{schema}}$, none of the literal determinant-bin label, canonical zero-mode record, physical group, or downstream shift lineage descends through π .*

Proof. The cut schema contains no crosswalk to the canonical parent $t = (\alpha, \gamma, j)$ of TPC-95. For example, with the same forgotten cut record and $h_0 = 2, j = 1$, two completion records may use slopes

$$(m_\alpha, m_\gamma) = (3, 1) \quad \text{or} \quad (5, 1).$$

Their target pairs are $(5, 3)$ and $(7, 3)$, and their normalized determinants are 2 and 4, respectively. Both values satisfy the displayed canonical integer formula, while the current cut fields distinguish neither completion.

Likewise the present schema does not constrain an outer affine key, its ordered coordinate, a physical group identifier, or a later stage target shift. Two completion records can agree after forgetting those fields and disagree on any one of them. Each label therefore fails the fiber-constancy condition of [theorem 3.1](#). This proves nonidentifiability relative to the current schema-completion class. It is not a mathematical impossibility statement on the unknown actual carrier: proving that stronger statement would require two completions satisfying every actual-support constraint. A future theorem-backed L_X can restrict the completion class and resolve the present ambiguity. $\square$

Remark 3.3 (Why this is a scoped stop). [Proposition 3.2](#) stops the implementation that derives the four downstream maps from the present JSON labels alone. It does not stop a new analytic or algebraic construction of L_X , and it does not show that the complete frontier scalar is large.

4 What the cut shift field does prove

Let $P_{h_0}^{\text{cut}}$ be the coordinate selector on cut rows whose stored shift is the prescribed h_0 .

Theorem 4.1 (Exact cut-stage selector). *On the complete TPC-136 cut archive,*

$$\boxed{P_{h_0}^{\text{cut}} = I.}$$

This is a literal L1 statement. It is not the downstream selector of TPC-125.

Proof. TPC-136 proves metadata intertwining for h_0 from every native column to every nonzero cut edge. The native packet is fixed at one prescribed h_0 , so every cut row is selected. $\square$

A downstream selector requires shift tags on the source and target of every later stage and an exact proof that those stages intertwine the selectors [\[1, 2\]](#). Copying h_0 into the cut metadata cannot certify a stage that has not been archived.

5 Executable field obligations

The standard-library program `experiments/tpc143_frontier_lift_audit.py` joins every TPC-136 nonsoft path to TPC-133 by exact packet scope and native tuple. It emits one obligation record per path. Downstream fields are recorded as `REQUIRED_MISSING`; no placeholder label is inserted.

Before forming that join, the program independently reruns the TPC-133–136 deterministic generators and compares their logical UTF-8 text under the canonical LF convention. The older certificates bind some Windows CRLF byte serializations, so their raw hashes are reported explicitly as `LEGACY_RAW_HASH_STALE`; they are not silently accepted as current hashes. The new `CANONICAL_UTF8_LF_V2` lock passes only after the generated atoms, paths, frontier manifest, cut paths, downstream-map manifest, and all certificate fields other than those diagnosed EOL-only locks agree.

For future affine-correlation consumers, each occurrence contract also reserves typed fields for the two affine forms, determinant, coefficient height, residue class, ordered interval, periodic weight, coprimality/squarefree/content/prefix masks, coefficient ℓ^1 -mass, window identifier and endpoint-ledger token. These are obligations, not values inferred from the cut tuple.

The committed regression has

$$\#C_X^{\text{ns}} = 2988, \quad \#C_X^{\text{ETO}} = 0, \quad \#C_X^{\text{FUM}} = 2988.$$

This census describes one finite sample at $X = 512$. Since the theoretical domain is the union $\text{ETO} \sqcup \text{FUM}$, the empty sample ETO set is never accepted as proof that the eligible tail is absent at growing scales.

Node	Status	Meaning
H1.cut_shift_selector	PROVED _{L1}	$P_{h_0}^{\text{cut}} = I$.
H1.frontier_occurrence_lift	NOT-TESTABLE	No conservative cut-to-occurrence matrix is archived.
$Q_D, Q_Z, G, P_{h_0}^{\text{down}}$	NOT-TESTABLE	Their literal source fields and map edges are absent.
Current-schema-only derivation	STOP-DECLARED-ROUTE	Stopped only by the scoped descent obstruction.

The audit rejects a missing or duplicated path, a source-terminal relabeling, a frontier-only domain, a downstream-field promotion, a false arithmetic-nonzero or L2 promotion, a fabricated occurrence edge, and promotion of $P_{h_0}^{\text{cut}}$ to a downstream map. Default mode writes deterministic artifacts; `-check` is read-only.

6 Claim boundary and next object

Theorem 6.1 (Audited occurrence-lift verdict). *For the current TPC-133–136 source artifacts,*

$$\boxed{\text{H1.frontier_occurrence_lift} = \text{NOT-TESTABLE.}}$$

The current schema alone cannot certify the four downstream labels, while the selected occurrence-augmented route remains open.

This paper proves no frontier $o(X)$ estimate, actual-carrier support theorem, determinant or zero-mode exponent, physical cover, downstream localization bound, endpoint below $1/400$, parity advance, prime-pair lower bound, or twin-prime theorem. Its positive content is an exact cut-stage selector and a necessary and sufficient field-descent test. The first object to build is the row-separated matrix L_X , not a collection of labels detached from their occurrence lineage.

References

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